

Multi-Cloud Disaster Recovery System

# Product Requirements Document

## Zero-Downtime Failover Across Cloud Regions

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Industry Skills: AWS - Azure - Disaster Recovery - Automation - High Availability

# 1. Executive Summary

The Multi-Cloud Disaster Recovery (DR) System is designed to guarantee business continuity for a company's critical applications and data in the event that its primary cloud region becomes unavailable due to outages, natural disasters, network failures, or provider-level incidents. The system provisions infrastructure across multiple regions (and optionally multiple cloud providers - AWS and Azure), continuously replicates data, monitors the health of all environments, and automatically fails over traffic and workloads to a healthy region with minimal to zero perceived downtime for end users.

## 2. Problem Statement

Modern businesses depend on always-on digital services. A single-region cloud deployment creates a single point of failure: a regional outage, misconfiguration, or disaster can take an entire application offline, causing revenue loss, SLA breaches, and reputational damage. Manual disaster recovery processes are slow, error-prone, and cannot meet modern Recovery Time Objective (RTO) and Recovery Point Objective (RPO) expectations.

## 3. Goals and Objectives

- Achieve near-zero downtime (target RTO under 5 minutes) during a primary region failure.
- Minimize data loss (target RPO under 1 minute) through continuous/near-real-time replication.
- Automate detection of regional failure and the failover decision, removing manual intervention from the critical path.
- Provide a single-pane-of-glass health monitoring dashboard across all regions/providers.
- Support automated, verifiable backups with defined retention and restore testing.
- Enable safe, automated failback to the primary region once it recovers.
- Design the system to be cloud-agnostic where feasible, supporting AWS as primary and Azure as secondary (or vice versa).

## 4. Scope

### 4.1 In Scope

- Multi-region active-passive (and optionally active-active) deployment architecture.
- Automated database replication (synchronous/asynchronous) between regions.
- Automated, scheduled and event-triggered backups with lifecycle management.
- Health monitoring and alerting across compute, database, network, and application layers.
- Automated failover orchestration (DNS/traffic shifting, promoting standby DB, scaling standby compute).
- Recovery orchestration workflows including failback procedures.
- Runbooks, dashboards, and reporting for DR readiness and incident history.

### 4.2 Out of Scope

- Application feature development unrelated to DR/HA.
- On-premises data center DR (cloud-to-cloud/cloud-region only).
- Physical hardware procurement.
- Third-party SaaS dependency failover outside company-managed infrastructure.

## 5. Stakeholders

| Role                               | Responsibility                                                  |
|------------------------------------|-----------------------------------------------------------------|
| Engineering / DevOps Team          | Build and maintain DR infrastructure, automation, and runbooks  |
| Site Reliability Engineering (SRE) | Own monitoring, alerting, and incident response                 |
| Product / Business Owner           | Define RTO/RPO targets and acceptable risk                      |
| Security & Compliance              | Ensure DR meets regulatory and data-residency requirements      |
| End Customers                      | Consumers of the application; impacted by downtime or data loss |

## 6. Functional Requirements

| ID    | Requirement                                                                                                                                | Priority |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-1  | System shall deploy application infrastructure identically across at least two cloud regions.                                              | Must     |
| FR-2  | System shall replicate the primary database to a standby database in the secondary region continuously.                                    | Must     |
| FR-3  | System shall perform automated, scheduled backups of databases and critical storage.                                                       | Must     |
| FR-4  | System shall continuously monitor the health of endpoints, services, and infrastructure in all regions.                                    | Must     |
| FR-5  | System shall automatically detect a primary region failure using multi-signal health checks.                                               | Must     |
| FR-6  | System shall automatically trigger failover: promote standby database, redirect traffic (DNS/load balancer), and scale up standby compute. | Must     |
| FR-7  | System shall notify designated stakeholders of failover events in real time.                                                               | Must     |
| FR-8  | System shall support automated or semi-automated failback once the primary region is confirmed healthy.                                    | Should   |
| FR-9  | System shall log every DR event with timestamps for audit and post-incident review.                                                        | Must     |
| FR-10 | System shall provide a dashboard showing real-time region health and replication lag.                                                      | Should   |

| ID    | Requirement                                                                         | Priority |
|-------|-------------------------------------------------------------------------------------|----------|
| FR-11 | System shall support manual override to trigger or abort a failover.                | Should   |
| FR-12 | System shall periodically test backup restore integrity (automated restore drills). | Could    |

## 7. Non-Functional Requirements

| Category                       | Requirement                                                                          |
|--------------------------------|--------------------------------------------------------------------------------------|
| Availability                   | Target 99.95%+ application availability (multi-region)                               |
| RTO (Recovery Time Objective)  | < 5 minutes for automated failover                                                   |
| RPO (Recovery Point Objective) | < 1 minute of data loss for critical databases                                       |
| Scalability                    | Standby region must scale to handle full production load within failover window      |
| Security                       | Encrypted data in transit and at rest; least-privilege IAM across regions/clouds     |
| Observability                  | End-to-end health metrics, logs, and traces across all regions                       |
| Cost Efficiency                | Standby resources sized/scheduled to balance cost vs. readiness (warm standby model) |
| Compliance                     | Meet data residency and audit logging requirements applicable to the business        |

## 8. Success Metrics / KPIs

- Measured RTO during simulated failover drills consistently under 5 minutes.
- Measured RPO (replication lag at time of failover) consistently under 1 minute.
- 100% of scheduled backups completed successfully with verified restorability.
- Zero manual steps required for a standard single-region failure scenario.
- Mean Time to Detect (MTTD) region failure under 60 seconds.

## 9. Assumptions and Constraints

- Company has active accounts/subscriptions on AWS and Azure (or two AWS regions as a minimum viable configuration).
- Application is largely stateless at the compute tier, allowing horizontal scaling of standby infrastructure.
- Database technology chosen supports cross-region or cross-cloud replication.
- Budget allows for a warm-standby footprint in the secondary region (not cold-only).

## 10. Risks

| Risk                                                     | Impact | Mitigation                                                                                           |
|----------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------|
| Split-brain during failover (both regions accept writes) | High   | Automated write-lock on old primary; quorum-based promotion; health-check consensus before promotion |
| Replication lag causing data loss                        | Medium | Near-real-time replication, lag alerting, RPO monitoring                                             |
| False-positive failover (unnecessary switch)             | Medium | Multi-signal health checks with confirmation window before triggering failover                       |
| Cost overrun from always-on standby                      | Low    | Warm-standby auto-scaling; scale-down when idle                                                      |
| Incomplete/untested backups                              | High   | Scheduled automated restore drills and integrity checks                                              |